



SMRITI — System Architecture

Tri-modal knowledge fabric over a typed industrial graph · 5 agents · runs fully local (no cloud)

SOURCES

the whole estate

P&ID drawings

Work orders (CMMS)

SOPs · rev chains

Inspections

Incidents / near-miss

OEM manuals · scanned

Permits to work

Email

OISD / Factories Act / PESO · sourced

▼ **Ingestion (Agent 1)** — per-type parsers · LLM entity/relation extraction vs typed schema · entity resolution

Continuous intake — structured forms · bulk CMMS CSV/JSON · auto-classifying upload (same graph writes as batch)

KNOWLEDGE STORE

typed + provenanced

Industrial Knowledge Graph — NetworkX, typed ontology, provenance on every node/edge, SUPERSEDES version chains

Qdrant · text (dense bge-small + sparse BM25)

Qdrant · visual (colSmol-256M multivector, MaxSim)

DrawingRegion index — pixels $\rightleftharpoons$ graph nodes

▼ **Retrieval fabric** — intent router (haiku) → **text || graph || visual** fan-out → RRF fusion → cross-encoder rerank → provenance + drawing overlays

AGENTS

evidence-constrained, all cited

2 · Expert Copilot — streaming, multi-turn

3 · Diagnostics / RCA — graph-native

4 · Compliance — clause $\rightleftharpoons$ record

5 · Lessons & Proactive Warnings

▼ **Presentation** — vanilla-JS PWA · streaming chat with [c#] citations · Drawing Viewer w/ live overlays + feed-path trace · reasoning-trace panel · dashboards

USER SURFACES

Ask

Assets

Diagnostics

Warnings

Compliance

Evaluation

Add data — live intake

One query, end to end: user asks → router classifies intent & tags → tri-modal fan-out → merge + rerank → provenance attached (doc, page, bbox, graph nodes) → answer streams with inline [c#] citations → if a drawing is in evidence, the viewer renders it with regions highlighted and the feed path traced → every step logged to the reasoning trace **and** the eval harness.

Trust as a feature

Every sentence cites; citations click through to the exact page/region; "insufficient evidence" is a first-class answer; faithfulness is measured against a vanilla-RAG baseline on a 33-question golden set (LLM-judged).

Built to run for years

New records enter through typed forms / bulk CMMS export / auto-classified upload and write the **same** typed :rec nodes as the batch build — so a new incident is instantly in the timeline and a new permit fires warnings live, stamped manual_intake · confidence 1.0 for audit.